Unit-4

ADVANCED TECHNIQUES OF PARALLELISM

- Parallelism is a process where large compute problems are broken down into smaller problems that can be solved simultaneously by multiple processors.
- It is the use of multiple processing elements simultaneously for solving any problem.
- Problems are broken down into instructions and are solved concurrently as each resource that has been applied to work is working at the same time.

Advantages

- It saves time and money as many resources working together will reduce the time and cut potential costs.
- It can be impractical to solve larger problems on Serial Computing.
- It can take advantage of non-local resources when the local resources are finite.
- Serial Computing 'wastes' the potential computing power, thus Parallel Computing makes better work of the hardware.

Types of Parallelism

- **Bit-level parallelism**

- It is the form of parallel computing which is based on the increasing processor's size. It reduces the number of instructions that the system must execute in order to perform a task on large-sized data.

- **Instruction-level parallelism**

- A processor can only address less than one instruction for each clock cycle phase. these instructions can be re-ordered and grouped which are later on executed concurrently without affecting the result of the program. this is called instruction-level parallelism.

- **Task Parallelism**

- Task parallelism employs the decomposition of a task into subtasks and then allocating each of the subtasks for execution. The processors perform the execution of sub-tasks concurrently.

- **Data-level parallelism (DLP)**

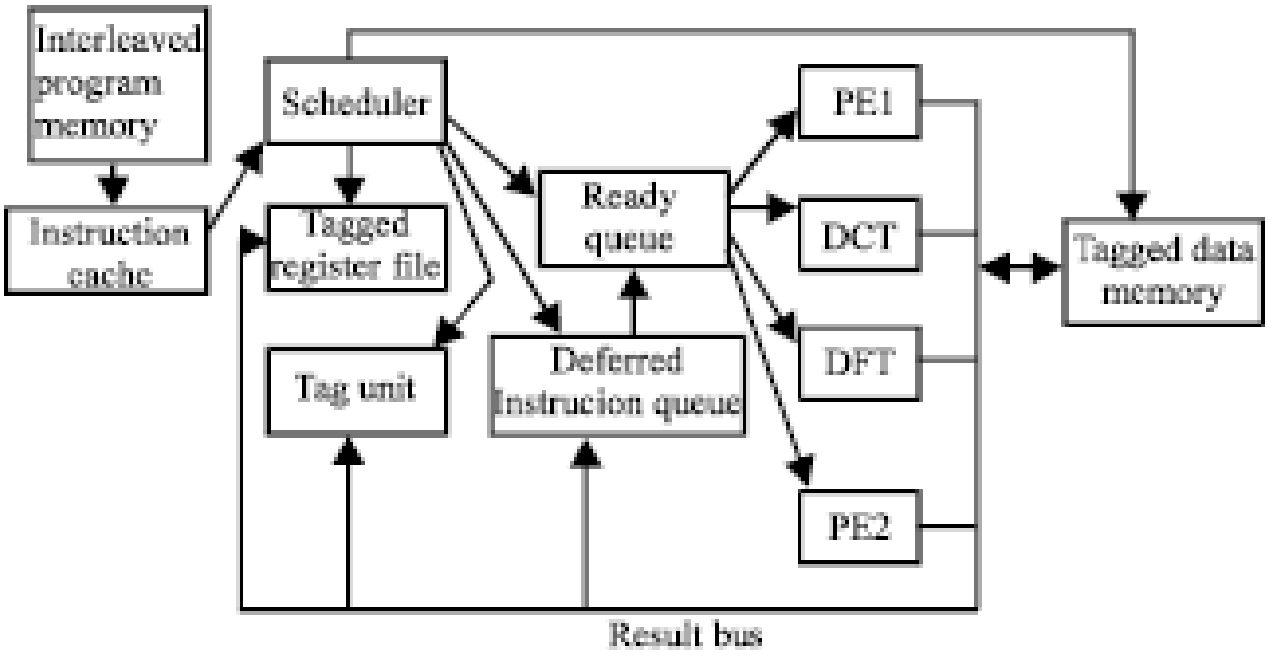
- It uses non-regular data manipulation patterns.

Instruction Level Parallelism (ILP)

- Instruction-Level Parallelism (ILP) refers to the capability of a processor to execute multiple instructions at the same time.
- Instead of running each instruction strictly one after another, ILP uses hardware and compiler techniques to overlap instruction execution wherever dependencies allow.
- Identifies independent instructions and runs them in parallel.

- Works within a single processor, not across multiple cores.
- Basis of modern CPUs with organised instruction scheduling.

Architecture of ILP



CYCLE	OPERATION
1	y1 = x1*1010
2	nop
3	nop
4	y2 = x2*1100
5	nop
6	nop
7	z1 = y1+0010
8	z2 = y2+0101
9	t1 = t1+1
10	p = q*1000
11	clr = clr+0010
12	r = r+0001

Fig. a

CYCLE	INT ALU	INT ALU	FLOAT ALU	FLOAT ALU
1	t1 = t1+1	clr = clr+0010	y1 = x1*1010	y2 = x2*1100
2	r = r+0001		p = q*1000	
3	nop			
4	z1 = y1+0010	z2 = y2+0101		

Fig. b

Advantages

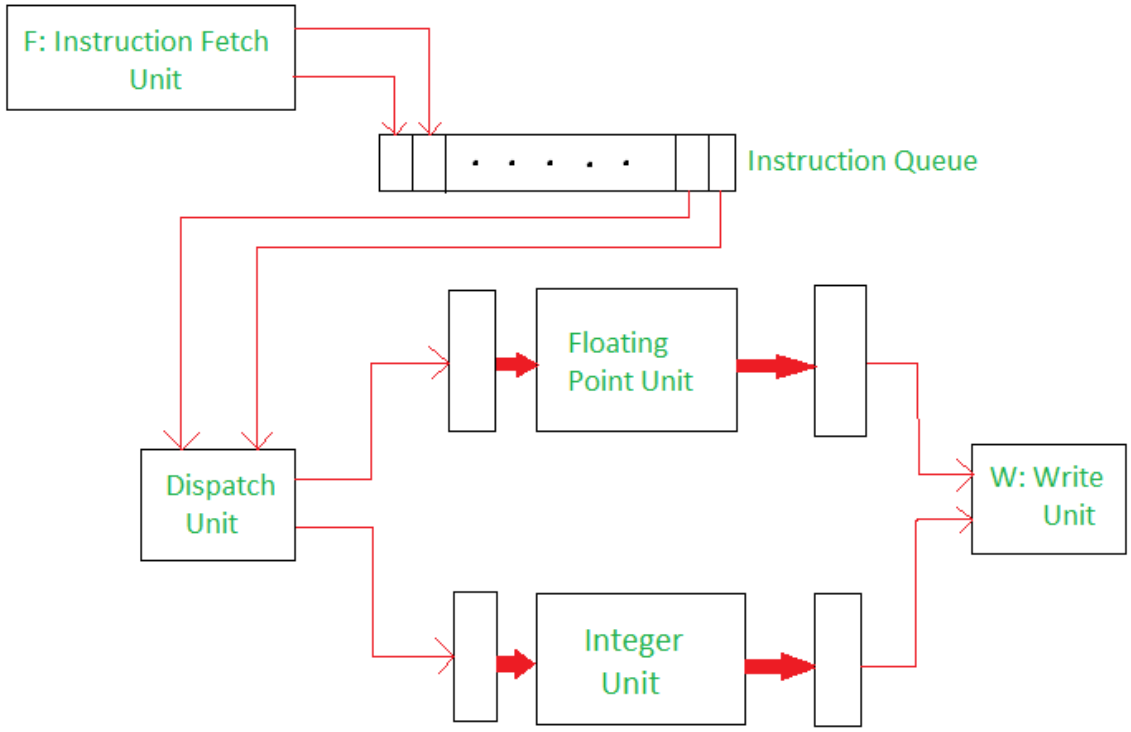
- Improved Performance
- Efficient Resource Utilization
- Reduced Instruction Dependency
- Increased Throughput

Disadvantages

- Increased Complexity
- Instruction Overhead
- Data Dependency
- Reduced Energy Efficiency

Superscalar Architecture

- A more aggressive approach is to equip the processor with multiple processing units to handle several instructions in parallel in each processing stage.
- With this arrangement, several instructions start execution in the same clock cycle and the process is said to use multiple issue.
- Such processors are capable of achieving an instruction execution throughput of more than one instruction per cycle is as **'Superscalar Processors'.**



Processor with Two Execution Units

- In the above diagram, there is a processor with two execution units.
- one for integer and one for floating point operations.
- The instruction fetch unit is capable of reading the instructions at a time and storing them in the instruction queue.

- In each cycle, the dispatch unit retrieves and decodes up to two instructions from the front of the queue.
- If there is one integer, one floating point instruction and no hazards, both the instructions are dispatched in the same clock cycle.

Advantages

- The compiler can avoid many hazards through judicious selection and ordering of instructions.
- The compiler should strive to interleave floating point and integer instructions. This would enable the dispatch unit to keep both the integer and floating point units busy most of the time.
- In general, high performance is achieved if the compiler is able to arrange program instructions to take maximum advantage of the available hardware units.

Disadvantages

- Due to this type of architecture, problem in scheduling can occur.

Flynn's Classifications

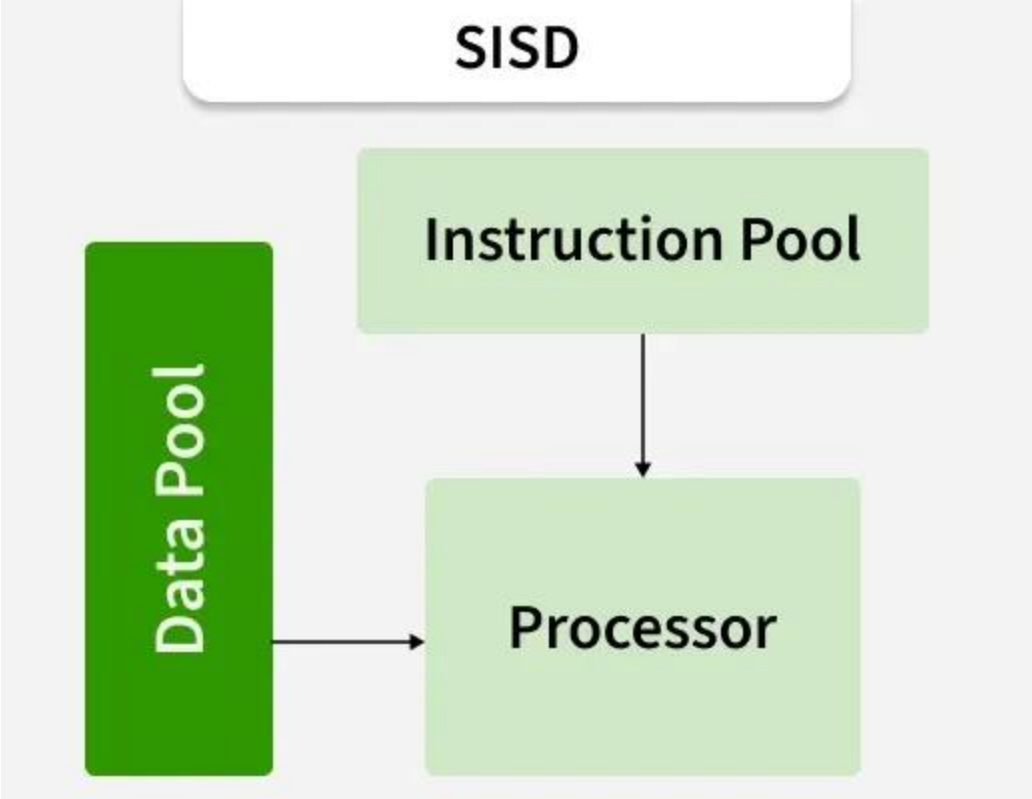
- It classifies computer architectures according to how many instruction streams and data streams they can process simultaneously.

They are…

- **Single-Instruction stream Single-Data stream(SISD)**
- **Single-Instruction stream Multiple-Data stream (SIMD)**
- **Multiple-Instruction stream Single-Data stream (MISD)**
- **Multiple-Instruction stream Multiple-Data stream(MIMD)**

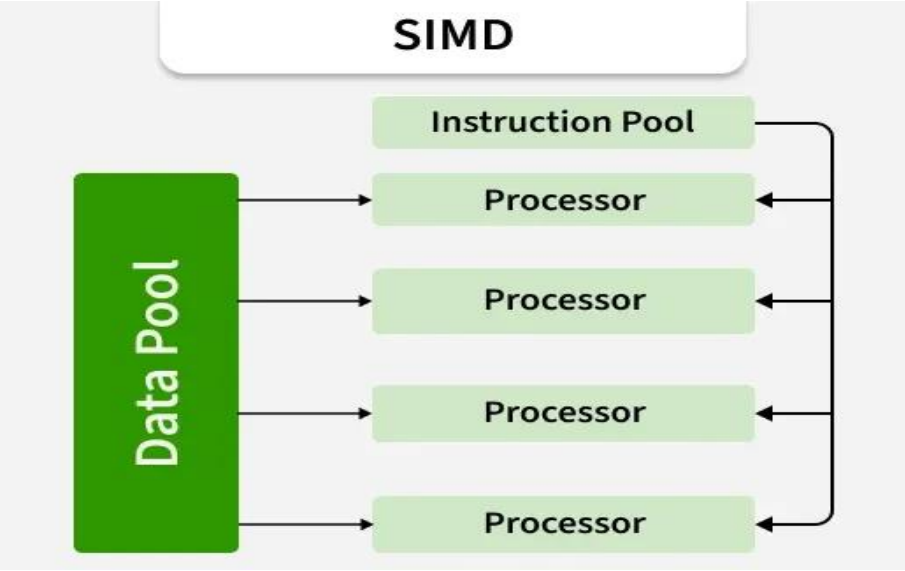
SISD

- Is a uniprocessor machine which is capable of executing a single instruction, operating on a single data stream.
- Here machine instructions are processed in a sequential manner, and computers adopting this model are popularly called sequential computers.
- Most conventional computers have SISD architecture.
- All the instructions and data to be processed have to be stored in primary memory.



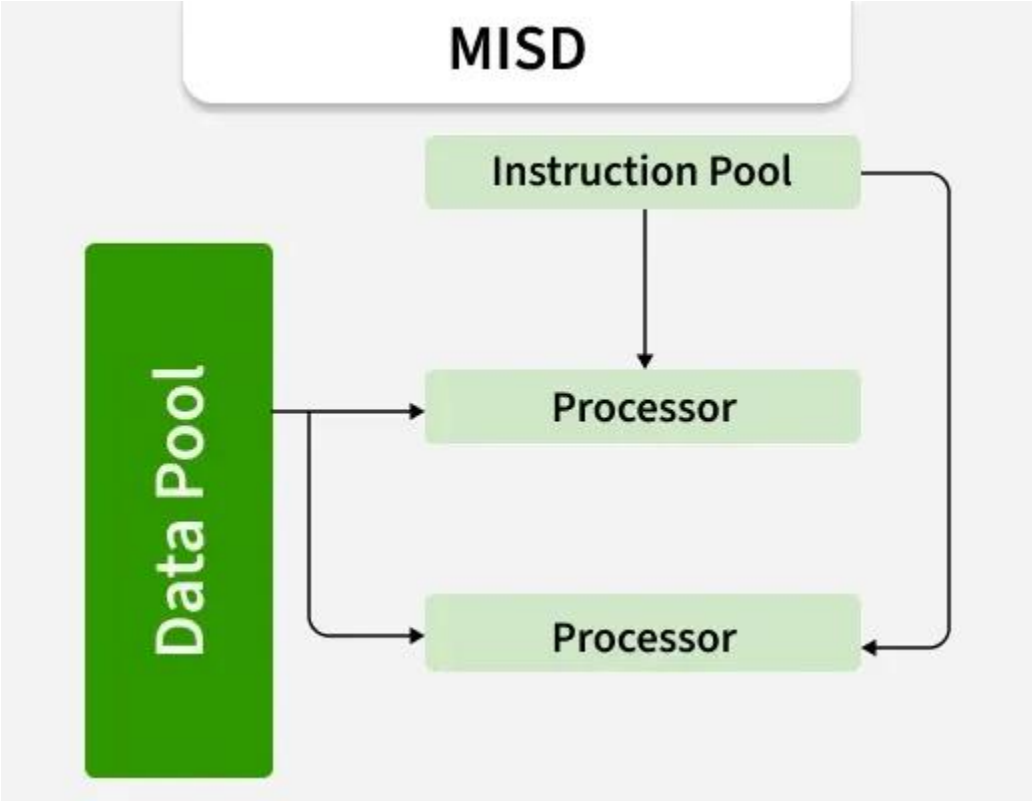
SIMD

- Is a multiprocessor machine capable of executing the same instruction on all the CPUs but operating on different data streams.
- Machines based on an SIMD model are well suited to scientific computing since they involve lots of vector and matrix operations.



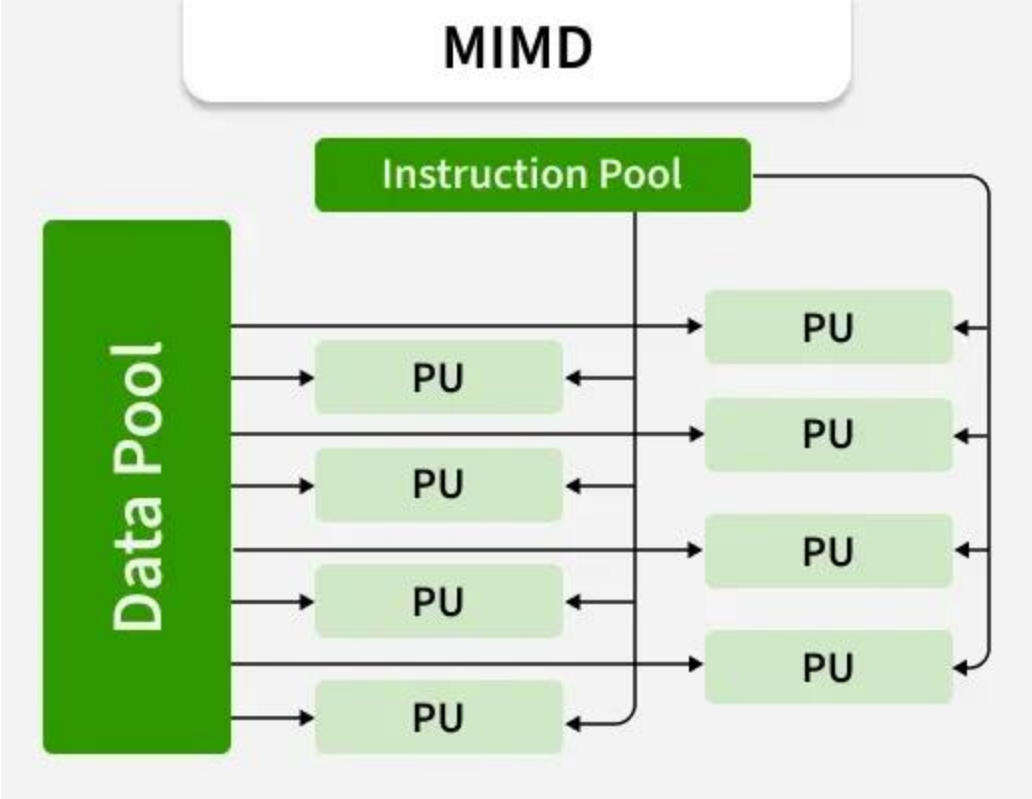
MISD

- is a multiprocessor machine capable of executing different instructions on different PEs but all of them operating on the same dataset.



MIMD

- Is a multiprocessor machine which is capable of executing multiple instructions on multiple data sets.
- Each PE in the MIMD model has separate instruction and data streams.
- Therefore machines built using this model are capable to any kind of application.

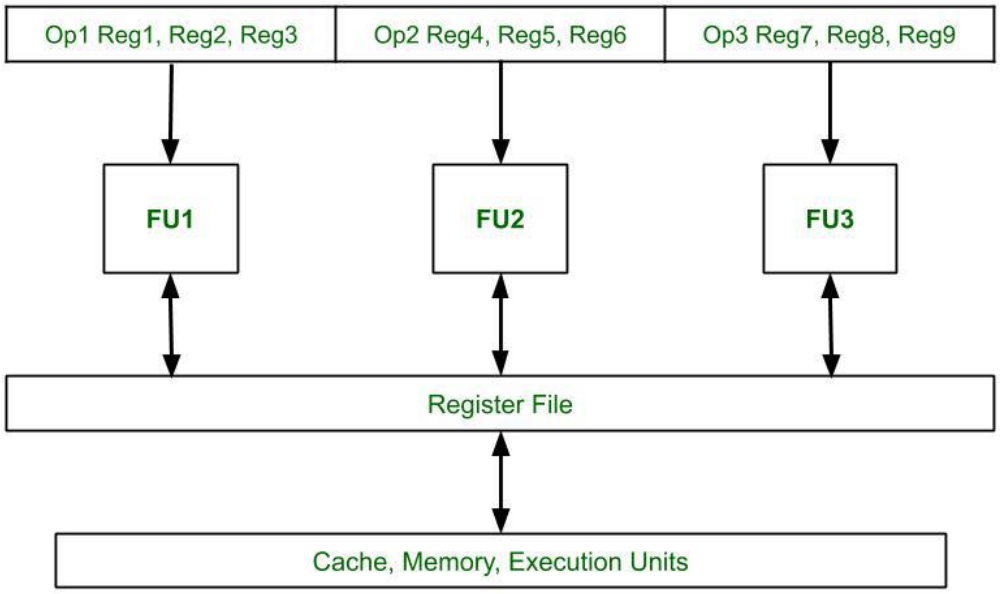


- MIMD machines are classified into **shared-memory** and **distributed-memory** models depending on how processors connect to memory.
- In a **shared-memory** MIMD system all processors use the same global memory, and communication happens through it.
- In a **distributed-memory** MIMD system each processor has its own local memory.
- They communicate through an interconnection network (like tree or mesh).
- For real-world use, distributed-memory MIMD is generally considered.

Very Long Instruction Word (VLIW) Processor

- Is a type of processor architecture designed to execute multiple operations in a single instruction cycle
- VLIW shifts this responsibility to the compiler, which packs independent instructions into one long instruction word.
- **Execution units work simultaneously on different operations.**
- **Reduces the need for complex runtime scheduling.**
- **Improves instruction-level parallelism.**

Architecture



- Here the performance of the processor is improved by using either of the following methods.
- **pipelining -** break the instruction into subparts
- **superscalar processor -** independently execute the instructions in different parts of the processor
- **out-of-order-execution -** execute orders differently to the program

Features

- The processors in this architecture have multiple functional units, fetch from the Instruction cache that have the Very Long Instruction Word.
- Multiple independent operations are grouped together in a single VLIW Instruction. They are initialized in the same clock cycle.
- Each operation is assigned an independent functional unit.
- All the functional units share a common register file.
- Instruction words length is 64-1024 bits depending on the number of execution unit and the code length required to control each unit.
- Instruction scheduling and parallel dispatch of the word is done statically by the compiler.
- The compiler checks for dependencies before scheduling parallel execution of the instructions.

Applications of VLIW Architecture

- **Digital signal processing (DSP**): VLIW processors are well-suited for DSP applications because of their ability to perform multiple operations in parallel.DSP applications require high computational power and often involve multiple parallel data streams, which VLIW processors can handle efficiently.
- **Multimedia processing**: VLIW processors are also used for multimedia applications such as video and audio processing, where high throughput and parallelism are required.
- **Scientific computing**: VLIW processors can be used for scientific computing applications, where high-performance computing is required to solve complex numerical problems.
- **Embedded systems:** VLIW processors are used in many embedded systems, such as automotive control systems, medical devices, and industrial automation equipment. These systems require high-performance processors that can execute multiple instructions in parallel while consuming minimal power.